

**SUPPLEMENTARY MATERIAL: How Much Replication Is Enough? Balancing Novelty
and Replication in a Truth-Driven Evolutionary Simulation**

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Focal sweep stability

In the main manuscript, we report a parameter sweep over our three focal parameters: base null rate, original study sample size, and publication bias strength. This sweep consisted of 10,000 simulations, drawn as 5 independent Latin hypercube samples of 2,000 parameter combinations each. Here, we assess the stability of the resulting marginal patterns by plotting each of these samples separately, in the same format as the marginal effects shown in the main manuscript, along with the combined results of all simulations.

These repetitions are not Monte Carlo simulations in the strict sense: they do not rerun identical parameter combinations with different random seeds. Rather, they use different samples of the parameter space to assess whether the observed patterns are stable across different samples of the same size. As shown in Figure 1, the marginal patterns are highly consistent across samples, giving us confidence that the main results are not an artefact of a particular parameter sweep.

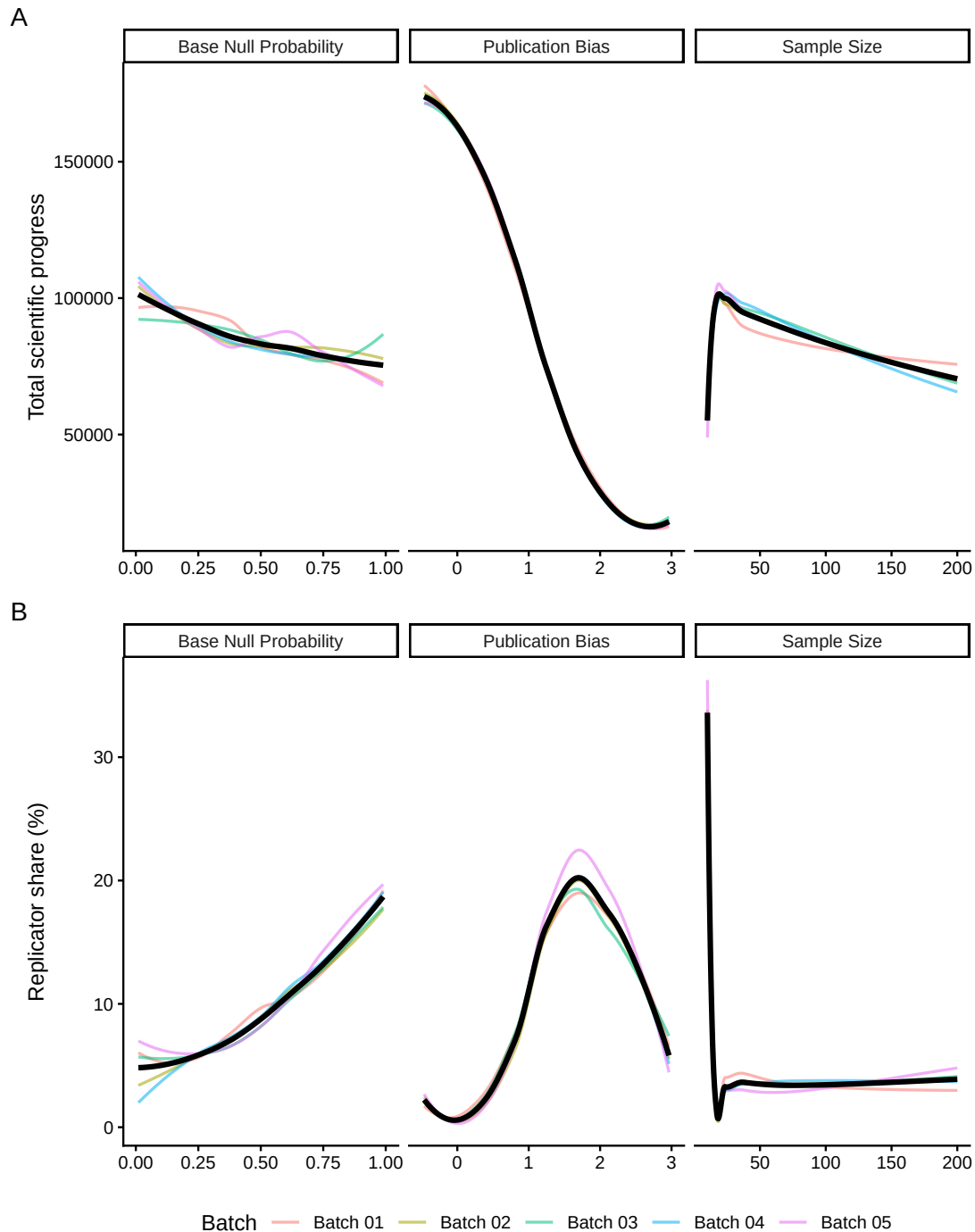
Non-focal parameter sensitivity

We also examined whether the focal sweep patterns were robust to alternative values of the non-focal parameters, which are held constant in the main manuscript analyses. Each scenario presented here is intended to represent a plausible alternative modelling assumption, by changing one or more non-focal parameters relative to the baseline configuration used in the focal sweep. Each scenario consisted of 2,000 simulations, using the same Latin hypercube sample of the focal parameter space as one batch of the main sweep. The full set of scenarios and parameter values is presented in Table 1. In Figure 2, we plot the sweep results for each scenario alongside the focal sweep results for comparison.

Overall, the scenario results were broadly consistent with the focal sweep. Most deviations were intercept shifts rather than qualitative changes in the shape of the marginal patterns. For total scientific progress (Panel A of Figure 2), the ‘large effects’ scenario produced higher overall progress. This is expected because larger true effects can generate larger observed effects, allowing individual studies to contribute more truthful information gain to the system. By contrast, the ‘tight priors’ and ‘optimistic priors’ scenarios produced lower scientific progress,

Figure 1

Marginal effects of publication bias strength, base null rate, and original study sample size on (A) total scientific progress at the end of the simulation and (B) the replicator share in the final population, estimated separately in independent samples of the focal parameter space



Note. Black lines show marginal patterns pooled across all simulations; coloured lines show the corresponding pattern in each independent batch of simulations sampled from the focal parameter space. The x-axis represents the full possible range of each parameter as follows. Sample size: 10 to 200, base null rate: 0 to 1, publication bias: -0.5 to 3.

because the tighter priors reduced the opportunity for original studies to substantially update flat community prior beliefs. The ‘slow originals’ scenario also produced lower scientific progress, because fewer original studies can be completed within the same amount of simulation time.

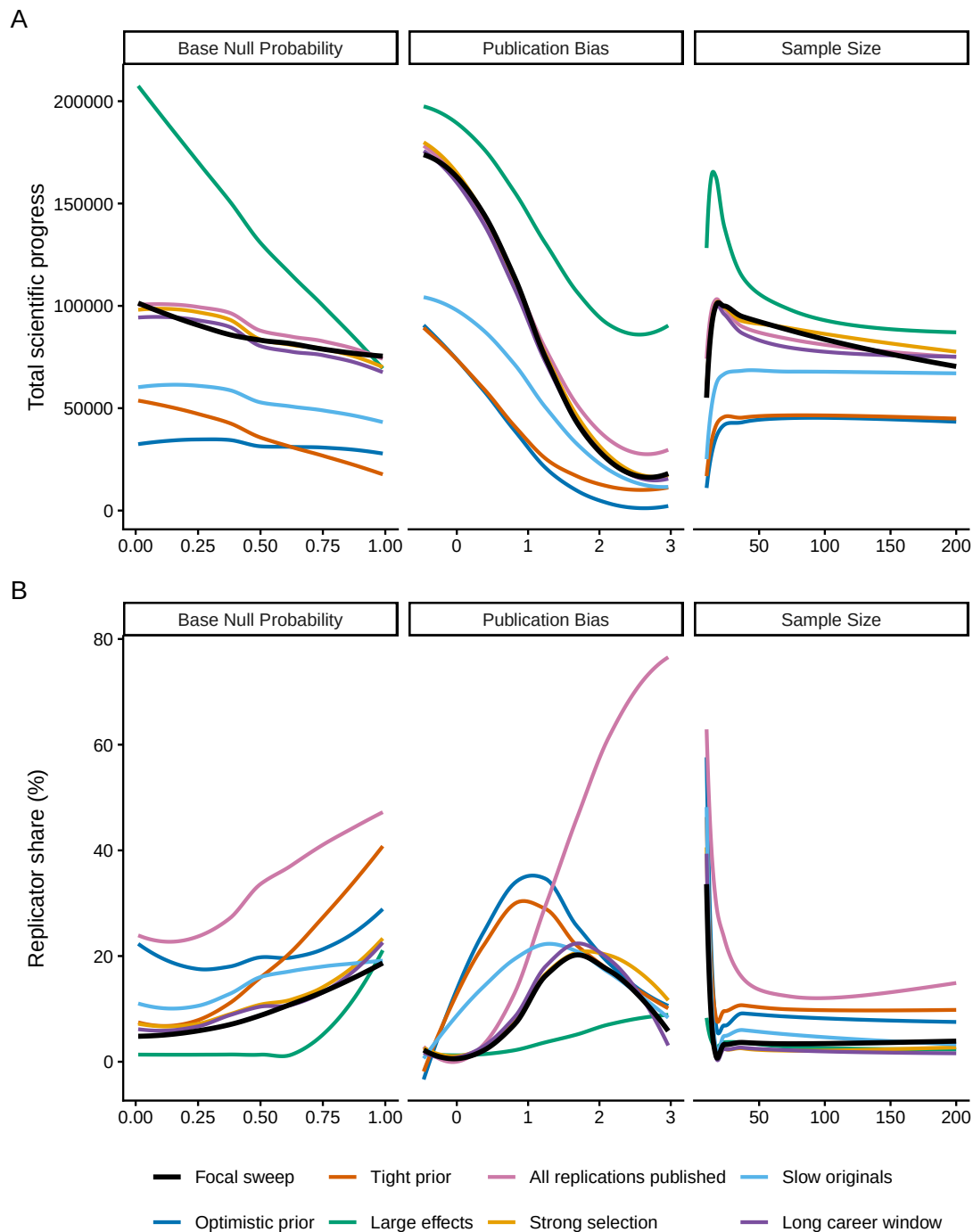
The clearest qualitative difference is seen in the relationship between publication bias and the evolved replicator share (Panel B of Figure 2). In the main manuscript analyses, increasing publication bias initially favoured replication, but this effect weakened once publication bias became so strong that replication was rarely published and therefore contributed little to researchers’ success. This pattern changed most clearly in the ‘all replications published’ scenario. In this scenario, replications being “not worth doing” no longer applies, because replication studies are always published and can therefore always contribute to community beliefs. Replicator agents also receive a hefty advantage because all of their completed studies enter the published record. As a result, this scenario produced higher overall replicator shares. For the ‘optimistic prior’ and ‘tight prior’ scenarios, the replicator share peaks at noticeably lower levels of publication bias. We believe this is because novelty is overall lower in these scenarios, so the level of publication bias at which replication is no longer worthwhile is reached sooner. Additionally, the ‘large effects’ scenario does not show this peak in replicator share, presumably because larger effects make results more robust overall. As publication bias rises, the replicator share is therefore lower overall and avoids the peak, producing a steadier increase.

Publication bias calibration

We aimed to calibrate the realistic condition as closely as possible to empirical patterns in the published literature. Publication bias is difficult to observe directly because it is characterised by studies that are never published (i.e., the file drawer problem). This is especially challenging for replication studies, which are already rare in the published literature and may be even more likely to remain invisible when they produce null or uninteresting results. We therefore calibrated the publication bias parameter indirectly, by sweeping the publication bias parameter and comparing emergent properties of the simulated published literature to empirical benchmarks.

Figure 2

Marginal effects of publication bias strength, base null rate, and original study sample size on (A) total scientific progress at the end of the simulation and (B) the replicator share in the final population, comparing the focal sweep with non-focal parameter scenarios



Note. Black lines represent the focal sweep; coloured lines represent each non-focal scenario. For the long career window scenario, total scientific progress is scaled to the standard simulation length equivalent for comparison with other scenarios. The x-axis represents the full possible range of each parameter as follows. Sample size: 10 to 200, base null rate: 0 to 1, publication bias: -0.5 to 3.

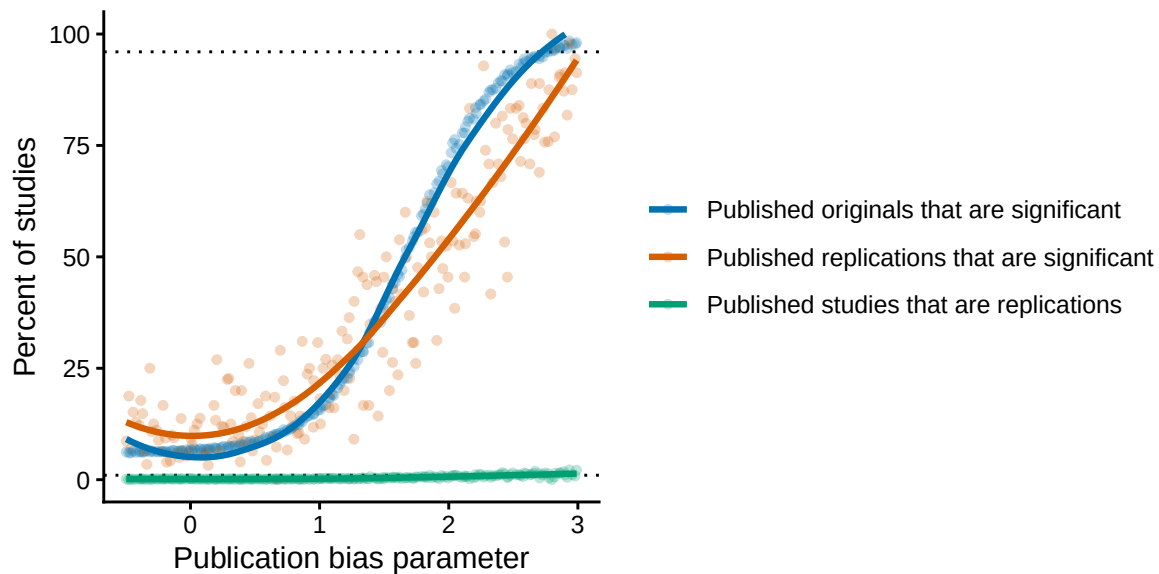
Specifically, we focused on two calibration targets: the proportion of published original studies that report statistically significant results, estimated at approximately 96% (Scheel et al., 2021), and the proportion of published studies that are replications, estimated at approximately 1% (Clarke et al., 2026; Hardwicke et al., 2020; Makel et al., 2012). We held all other parameters fixed while sweeping publication bias strength across 200 simulations and observing the resulting literature over 100 timesteps. We disabled career selection and mutation so that the researcher population did not evolve, and fixed the replicator share at 1% to anchor the supply of replication effort near the empirical prevalence of replications in published psychology literature. We set original study sample size to 20 per group, reflecting the lower end of historically typical sample sizes in psychology (Bakker et al., 2025; Fraley & Vazire, 2014; Marszalek et al., 2011), and set the base null rate to 0.9, corresponding to a highly exploratory research environment in which most true effects are null (Ioannidis, 2005).

The results of this calibration sweep are shown in Figure 3. Based on these patterns, publication bias values between 2 and 3 appear to provide the closest match to the empirical targets, and we therefore treat this range as a plausible setting for the realistic condition.

Extended methods

This extended methods section mirrors the main document methods while adding more detail to each of the submodel processes shown in Figure 4. All default and swept parameter values are shown in Table 2.

We investigated the optimal allocation of research effort between original research and replication using an agent-based model of cultural evolution in academia, implemented in R. The model allowed us to simulate how individual researcher behaviour, operating under selection pressure, shaped epistemic outcomes over time. Specifically, we wanted to know the replicator share that emerges under selection for truthful scientific progress. The replicator share is the percentage of active researchers with a replication research style during the final 50 timesteps of the simulation. Because researchers work continuously on one study at a time, this is also the share of researcher time - that is, of research effort - allocated to replication. Thus, we can use the

Figure 3*Composition of the simulated published literature across publication bias parameter values*

Note. Each point is one simulation. Horizontal dotted lines indicate empirical benchmarks used for calibration: 96% of published original studies reporting significant results and 1% of published studies being replications.

replicator share as a direct measure of how limited scientific resources are allocated between original and replication research. Within our model, selection is utopian: researchers' survival in academia relies on truth contribution, not the productivity- and novelty-based rewards that govern real academic careers. Researchers in our model neither p-hack nor fabricate data, thus observed effects depart from true values only through sampling variability - omitting other error sources that would likely raise the replicator shares we report. The academic system is otherwise intended to capture realistic aspects, including publication bias, so the replicator share that evolves can be read against plausible research conditions. We also varied three focal parameters across plausible ranges to assess their influence on the optimal allocation of research effort: the base null rate of effects, the sample size of original studies, and the strength of publication bias. The model was preregistered using an ODD (Overview, Design concepts, and Details) model description (Grimm et al., 2006), [available on Zenodo](#); deviations from the preregistration are listed at the end of the

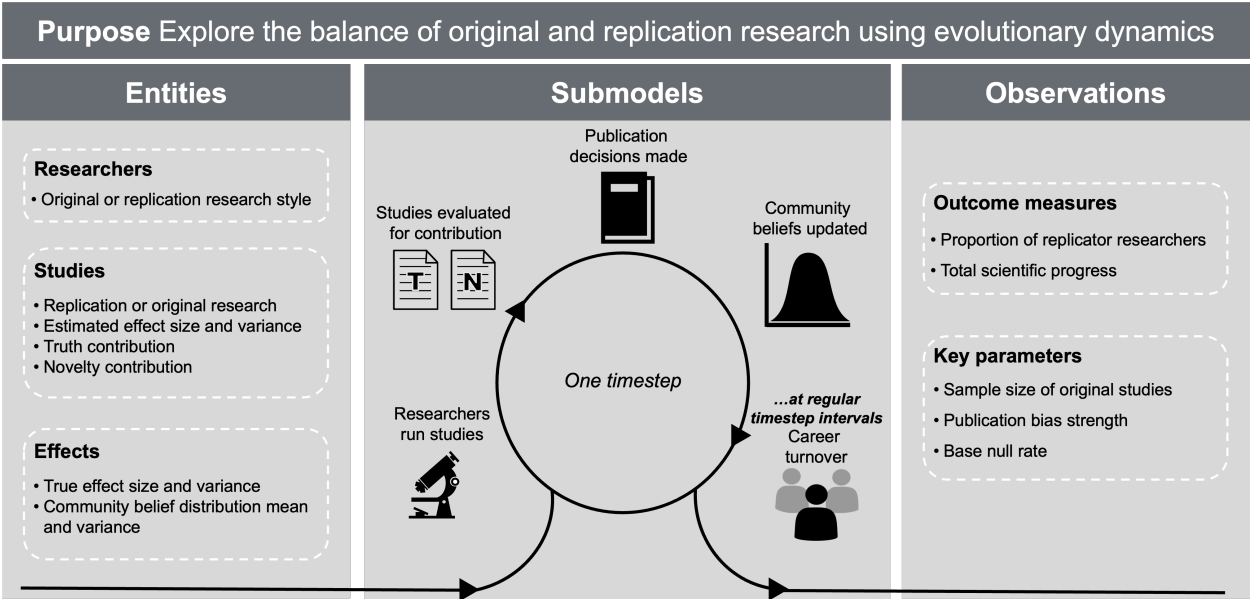
Methods.

Figure 4 provides a visual overview of the model based on the ODD protocol (Szangolies et al., 2024). At each timestep, agents conduct either original or replication studies according to their research style. Studies take multiple timesteps to complete, contingent on sample size (i.e., larger sample sizes take longer to complete). When a study is completed, it is assigned a novelty contribution score, representing change in community beliefs (i.e., information gain), and a truth contribution score, representing change in community beliefs relative to the true effect. Both contribution scores follow a Bayesian belief updating framework and are explained further below. Publication bias then determines which studies enter the published literature, based on their statistical significance and novelty contribution. Only published studies update the actual community belief distribution for the relevant effect. At regular timestep intervals, researchers enter a career turnover phase, in which their retention or exit in the academic system depend on the cumulative truth contribution of their published studies during the preceding phase. Exited researchers are replaced with new researchers who inherit their research style from the remaining researchers, with a small chance of mutation to capture imperfect transmission, individual variability, and innovation. At the end of each timestep, the model records the current system state, including the distribution of research styles within the population and the overall scientific progress (i.e., the cumulative truth contribution of all studies). The following sections describe the model initialisation and the submodels depicted in Figure 4 in more detail.

Initialisation

Each simulation begins with a population of 1,000 researchers who conduct original studies and have an “active” status (i.e. they are currently working in academia). A pool of 500,000 effects is generated to ensure that the supply of original effects is effectively inexhaustible; if this pool is depleted, more effects are added. The true effect size values are generated such that they have a specified probability of being 0, determined by the base null rate parameter. For non-null effects, the true effect size in the Cohen’s d metric is drawn from a normal distribution with a mean of 0.3 and a standard deviation of 0.1. Each effect is also

Figure 4
Model overview



Note. ODD = Overview, Design concepts, and Details. “T” and “N” indicate that studies were evaluated for their contribution to truth and novelty, respectively. Individual graphics were sourced from publicdomainvectors.org (CC0).

assigned a prior Bayesian community belief distribution, initialised as a normal distribution with a mean of 0 and a variance of 1, indicating that before any studies have been conducted the scientific community has no strong expectations about whether an effect exists or how large it might be. There are no existing studies at the beginning of a simulation.

Researchers run studies

Researcher time is the key limited resource in the model: only a fixed number of researchers are active at any time, each devotes their effort to a single study until it is finished, and there is no idle time between studies. Each timestep, researchers who have completed a study start a new one; researchers who are mid-project continue until completion or until a potential drop-out in the career turnover stage (see below). Original researchers always run original studies and replicator researchers always run replication studies.

For original studies, researchers randomly draw from a pool of effects that are not yet part of the published record and are not currently being investigated in another ongoing, publishable

project. This exclusion rule serves to maintain computational simplicity by preventing multiple “first” studies on the same effect from being launched in parallel. For replication studies, researchers can only select effects that have already been published at least once. The model prioritises effects with the fewest published replications (i.e. all original studies will be replicated once before any are chosen to be replicated a second time). If there are more replication attempts queued than there are eligible published effects, the excess replication attempts are reassigned to original studies for that timestep. Because the initial population consists entirely of researchers conducting original studies, this situation is uncommon: the early generations typically produce a large published literature before replicator agents become common in the population.

Sample size per group for original studies is fixed within each simulation and varied across simulations (from 10 to 200 per group). For replication studies, sample size is chosen dynamically to provide 80% power ($\alpha = .05$, one-sided) to detect the original reported effect when it was significant, or an effect of $d = 0.3$ when it was non-significant. The model also accounts for the time required to conduct research. Study duration increases by 0.1 timesteps for each observation per group, and original studies require one additional timestep for planning before they can be completed.

All studies are simulated as independent-samples t -tests, with sample size defined as the number of observations per group. When a study is conducted, its observed t statistic is drawn from a noncentral t distribution determined by the true effect size and the per-group sample size. This statistic is then converted into a standardised effect size estimate and corresponding standard error. The resulting p -values are calculated from the observed t statistic and degrees of freedom: original studies use two-sided tests, whereas replications use one-sided tests in the direction of the original estimate.

Studies evaluated for contribution

Each completed study is evaluated for its potential contribution, as if its estimate had already been used to update the community belief distribution about the relevant effect. However, the community belief distribution is only actually updated if the study is published. We

distinguish between two measures of scientific contribution. *Novelty contribution scores* capture how much the study would change community beliefs, regardless of whether the change is in the correct direction. Technically, it is the difference between the prior belief distribution and the posterior belief distribution that would result from incorporating the study estimate, measured using Kullback-Leibler divergence (D_{KL}); this is also commonly referred to as information gain, where larger values indicate a larger shift in beliefs. *Truth contribution scores* capture how much the study would change community beliefs toward the true effect size (known to us, the modellers). Truth contribution is measured by comparing how much weight the prior and posterior distributions place on the true value. Positive truth contributions mean the updated belief distribution places more weight on the truth than before the study, while negative values mean there is now less weight on the true value. Note that this framework also treats correctly learning that an effect is null as scientific progress, because reducing uncertainty around a true null increases the posterior density at the true value. To be clear, this is a utopian measure that agent-based modelling allows us the opportunity to know and utilise, as in reality we have no access to the true effect size. (And if we had, more research would be futile.) Let $p_{\text{prior}}(\theta)$ denote the existing community prior and $p_{\text{post}}(\theta)$ the posterior that would be obtained if the study were published. Studies are evaluated as such:

$$\text{Novelty contribution score} = D_{\text{KL}}(p_{\text{post}}(\theta) \parallel p_{\text{prior}}(\theta))$$

$$\text{Truth contribution score} = \log p_{\text{post}}(\theta_{\text{true}}) - \log p_{\text{prior}}(\theta_{\text{true}})$$

The truth contribution score is computationally equivalent to a Bayes Factor estimated by the Savage-Dickey density ratio. Both measures move beyond the frequentist significant/non-significant dichotomy and consider the continuous nature of a shift of beliefs. The absolute values of these scores do not have a straightforward interpretation and are most meaningful when compared across simulated studies. Because novelty and truth contribution are defined differently and operate on different scales, their values cannot be compared directly with

one another.

The novelty contribution is used as part of the publication-bias process while the truth contribution serves as the measure of researcher fitness: the cumulative truth contribution of a researcher's published studies within a career phase. When assessing the total scientific progress of a simulation, we sum across all effects that received at least one published update, comparing each effect's final posterior belief distribution to the uninformed prior from the start of the simulation.

Publication decisions made

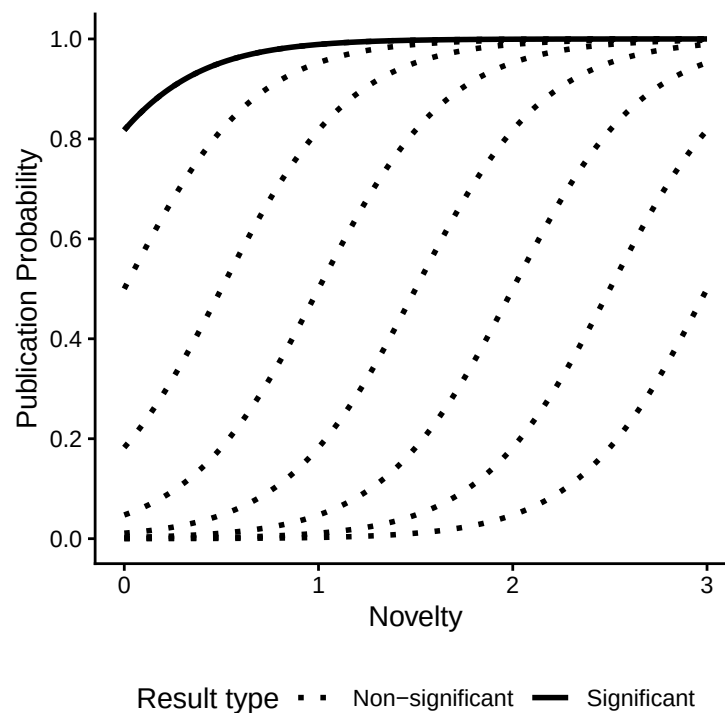
Publication bias operates on two criteria: the novelty contribution of the study and the statistical significance of its finding (p -value). Publication bias follows two axioms: first, given the same level of novelty, significant results are more likely to be published than non-significant ones; and second, given the same status of significance, novel findings are more likely to be published than less novel ones. These axioms together generate a consistent outcome hierarchy, ranging from the lowest publication probability for low-novelty, non-significant studies to the highest for high-novelty, significant studies. A study's probability of publication follows one of two logistic functions based on novelty and significance:

$$\Pr(\text{publish} \mid p < .05, N) = a_{\text{sig}} + \frac{1 - a_{\text{sig}}}{1 + \exp[-k_{\text{sig}}(N - m_{\text{sig}})]}$$

$$\Pr(\text{publish} \mid p \geq .05, N) = \frac{1}{1 + \exp[-k_{\text{nonsig}}(N - m_{\text{nonsig}})]}$$

where N is a study's novelty contribution, a_{sig} is the lower asymptote for significant results, m terms are logistic midpoints, and k terms are logistic steepness parameters.

Publication bias is operationalised through the m_{nonsig} term: higher values of publication bias mean a lower baseline probability that a non-significant study is published, and a higher novelty threshold for such a study to be published despite its non-significance (visualised in Figure 5). Note that the publication bias function applies both to original and replication studies in the focal simulations.

Figure 5*Publication bias functions*

Note. Multiple logistic curves for non-significant findings are presented to visualise the range of publication bias conditions, with the m_{nonsig} term ranging from -0.5 to 3. At -0.5, non-significant and significant findings have the same probability of publication for a given level of novelty, so the two curves overlap and only the significant curve is visible. At 3, the probability of publishing a non-significant finding is essentially zero unless novelty is extremely high.

Community beliefs updated

Once published, study outcomes are used to update the community's beliefs about the studied effect. Community beliefs are modelled as probability distributions and are updated using Bayesian inference, specifically normal-normal conjugacy, where the prior distribution is updated using the study's observed effect size and standard error to produce a posterior distribution (see equation below, where y denotes the study evidence used as a likelihood distribution). This posterior replaces the prior as the updated community belief distribution, allowing community

knowledge to accumulate across studies.

$$p_{\text{post}}(\theta | y) = p(y | \theta) p_{\text{prior}}(\theta)$$

Applying this approach to the real world, the simulated agents run perfect community-augmented meta-analyses on the entire published literature.

Career turnover

Career turnovers occur every 35 timesteps, allowing researchers to complete multiple studies under most sample-size conditions, while preserving the substantial time cost of very large studies. Original studies have a fixed planning cost of 1 timestep plus 0.1 timestep per per-group observation. At the lowest sample size of 10 per group, a study takes 2 timesteps, allowing 17 studies per career phase; at the highest sample size of 200 per group, a study takes 21 timesteps, allowing a minimum of one study per phase. At each career turnover phase, researchers are ranked by their truth-based fitness, defined as the cumulative truth contribution of their published studies during the previous phase (i.e., summed log-density gains at θ_{true} , as defined earlier). The lowest-ranked half of agents exit academia and are marked as inactive, while the remaining agents are promoted, meaning they are allowed to continue in the academic system. If ties occur, promotions among tied candidates are decided randomly. Vacant positions are then filled by new researchers, who inherit their research styles from existing values in the active academic population, with a small chance of mutation (2% probability of switching from one style to the other) to capture imperfect transmission, individual variability, and innovation. The active population therefore remains fixed at 1,000 while the replicator share evolves over time.

Deviations from preregistered model design

Several aspects of the model were refined during development. First, we now define truth contribution using the change in log density assigned to the true effect size, rather than the originally specified ‘verisimilitude’ measure, $\text{KL}(q \parallel \text{prior}) - \text{KL}(q \parallel \text{post})$ with q a very narrow normal distribution centred on the true effect. The two measures are highly correlated, but the

log-density measure is more direct, easier to interpret as the probability density at the true effect, and avoids introducing an arbitrary width to the truth distribution.

Second, our preregistration also included a condition in which researcher selection was based on novelty rather than truth. In most conditions, the novelty and truth selection results were qualitatively very similar, because greater unbiased information gain generally also moves the literature closer to the truth. The main exception was under strong publication bias, where the information gain entering the literature and shaping community beliefs was selected in a systematically biased way. In addition, the novelty selection condition effectively applied novelty pressure twice: once through the publication process, where more novel findings were more likely to be published, and again through career selection, where researchers were rewarded for producing novel contributions. This created some less interpretable dynamics that were not central to the main question of how much replication would evolve under truth selection. We thus excluded the novelty condition and the related corrective condition.

Third, we revised the role of publication bias in the model. Earlier descriptions implied that truth selection would be paired with no publication bias. However, our aim is to isolate the effect of changing selection on researchers while keeping the broader academic environment otherwise realistic. We therefore allow publication bias to remain present. Because publication bias is also swept as a focal parameter, the model still includes conditions with little or no publication bias, as described in the preregistration.

Fourth, we modelled researchers as either exclusively conducting original studies or exclusively conducting replications, rather than having a “replication probability” as preregistered. We adopted these dichotomous researcher types because the probabilistic determination of study types introduced noise that obscured the evolutionary dynamics.

Finally, we removed individual variation in researchers’ target power. Although this was included in an earlier version of the model, it added another adaptive mechanism that made the replication dynamics difficult to observe and interpret. We instead vary original study sample size as an environmental parameter, which keeps the model focused on the balance between original

and replication research while still capturing differences in study reliability.

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Table 1*Non-focal parameter sensitivity scenarios.*

Scenario	Baseline → scenario parameters	Description
Tight prior	Prior mean = 0 Prior variance = 1 → 0.25	The scientific community starts with a more informative (narrower) prior.
Optimistic prior	Prior mean = 0 → 0.3 Prior variance = 1 → 0.25	The scientific community starts with a more optimistic belief that effects are positive, and a narrower prior.
Large effects	Effect size mean = 0.3 → 0.8	True (non-null) effects are drawn from a distribution with a larger mean.
All replications published	All replications published = 0 → 1	Replication studies enter the published literature regardless of significance or novelty.
Strong selection	Career selection rate = 0.5 → 0.8	A larger share of researchers exit academia and are replaced at each career turnover phase.
Slow originals	Original study intercept = 1 → 5	Original studies have higher startup requirements; taking even longer to complete compared to replication studies.
Long career window	Career phase length = 35 → 70 Total timesteps = 1,000 → 2,000 Burn-in period = 35 → 70	Researchers have a longer time to run studies before being evaluated; other timestep parameters are scaled to match.

Table 2*Parameter values.*

Parameter name	Description	Default value (swept values)
Original study sample size	Per-group sample size for original studies (varied)	20 (10–200)
Publication bias strength (m_{nonsig})	Midpoint of the logistic publication function for non-significant results (varied)	3 (-0.5–3)
Base null rate	Probability that a true effect is null (varied)	0.9 (0–1)
Number of researchers	Active researchers in the population	1,000
Total timesteps	Length of each simulation	1,000
Career phase length	Timesteps between career turnover events	35
Burn-in period	Initial timesteps with originals only and no mutation	35
Study duration per observation	Timesteps added per per-group observation	0.1
Original study planning intercept	Fixed additional timesteps for original studies	1
Effect pool size	Number of true effects available to study	500,000
Non-null effect size mean	Mean of the true non-null effect size distribution (Cohen's d)	0.3
Non-null effect size SD	Standard deviation of the true non-null effect size distribution	0.1
Prior mean	Mean of the uninformed community prior	0

Parameter name	Description	Default value (swept values)
Prior variance	Variance of the uninformed community prior	1
Career selection rate	Proportion of researchers replaced at each career turnover	0.5
Mutation rate	Probability that a new researcher switches research style	0.02
Initial replicator share	Starting share of replicator researchers	0
Significant results lower asymptote (a_{sig})	Minimum publication probability for significant results	0
Significant results midpoint (m_{sig})	Novelty midpoint of the logistic publication function for significant results	-0.5
Significant results steepness (k_{sig})	Steepness of the logistic publication function for significant results	3
Non-significant results steepness (k_{nonsig})	Steepness of the logistic publication function for non-significant results	3
All replications published	Whether replications bypass publication bias (1 = always published)	0